

KALUGA, Russian Federation

WMO: 277030

Lat: **54.550N**Lon: **36.367E**Elev: **652**StdP: **14.35**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.9	-6.6	-18.0	2.1	-12.1	-11.1	3.1	-5.9	22.8	23.4	20.9	26.5	3.0	30	0.592

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.1	83.4	67.1	80.5	65.7	77.7	64.9	70.0	79.0	68.1	76.6	66.5	74.7	7.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.6	100.3	74.4	64.9	94.5	72.5	63.3	89.3	70.9	34.3	79.6	32.8	76.4	31.5	74.5	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation	n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max			Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4) 20.0	17.0	15.1		-19.8	87.7	6.5	3.5	-24.4	90.2	-28.2	92.3	-31.9	94.2	-36.6	96.8	
(5)			DB	-19.8	72.8	6.3	1.9	-24.3	74.1	-28.0	75.2	-31.6	76.3	-36.1	77.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.0	19.4	20.0	28.6	43.9	55.2	61.0	65.8	62.1	52.5	41.8	30.7	22.0	
	DBStd	18.90	11.90	12.28	8.66	8.98	8.26	6.48	5.32	5.57	7.27	7.80	9.77	11.96	
	HDD50	4505	950	840	665	226	45	3	0	0	56	272	580	868	
	HDD65	8540	1415	1260	1130	632	315	152	55	124	376	719	1029	1333	
	CDD50	1600	0	0	0	44	206	333	491	375	132	18	1	0	
	CDD65	161	0	0	0	0	11	33	81	34	2	0	0	0	
	CDH74	1569	0	0	0	8	166	359	665	340	31	0	0	0	
	CDH80	295	0	0	0	1	21	64	152	56	1	0	0	0	
Wind	WSAvg	7.2	8.0	8.1	8.4	7.7	6.9	6.6	5.9	5.8	6.2	7.2	7.8	8.0	
Precipitation	PrecAvg	26.0	1.9	1.4	1.5	1.4	2.4	3.1	3.2	2.3	2.6	2.7	2.0	1.7	
	PrecMax	33.5	2.8	2.6	2.8	2.5	4.7	5.6	5.9	6.4	7.0	7.4	4.2	3.2	
	PrecMin	16.3	0.8	0.4	0.6	0.6	0.6	1.2	0.2	0.1	0.9	0.5	0.1	0.6	
	PrecStd	4.4	0.6	0.6	0.7	0.6	1.2	1.3	1.5	1.7	1.6	1.7	0.9	0.7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.6	39.3	54.2	74.5	82.7	85.4	88.2	85.7	77.4	66.4	52.3	44.9	
		MCWB	36.6	36.3	43.9	55.9	63.7	66.9	68.1	68.0	61.4	55.6	49.4	43.3	
	2%	DB	36.5	37.3	47.8	68.9	79.1	81.8	84.2	80.9	73.6	60.9	48.2	39.5	
		MCWB	35.0	35.0	39.6	53.6	62.7	66.4	67.9	66.3	61.0	54.0	46.0	37.8	
	5%	DB	34.4	35.7	43.3	64.2	74.8	78.4	80.8	78.0	69.3	56.9	45.6	36.1	
		MCWB	33.2	33.7	37.3	51.1	60.2	64.7	67.2	65.1	59.5	52.0	43.7	34.9	
	10%	DB	33.1	34.4	39.5	59.7	70.6	74.8	77.6	74.5	64.7	53.3	42.3	34.2	
		MCWB	31.9	32.9	35.5	49.2	57.9	63.4	66.2	63.5	57.1	49.5	40.9	33.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.3	37.3	44.9	57.9	66.8	70.7	72.9	71.4	64.5	58.3	50.6	43.6
			MCDB	38.0	38.3	52.0	70.9	79.2	81.4	82.9	80.8	74.0	63.2	52.3	44.1
2%		WB	35.2	35.5	40.8	54.7	63.8	68.1	70.8	68.7	62.1	55.0	46.5	38.1	
		MCDB	36.0	36.7	46.3	65.3	74.9	77.8	80.0	77.2	70.2	59.3	47.7	39.1	
5%		WB	33.4	34.1	38.1	52.7	61.7	66.2	68.9	66.7	60.1	52.3	43.8	35.2	
		MCDB	34.2	35.2	41.8	62.3	71.9	75.6	77.2	74.8	67.0	56.2	45.2	36.0	
10%		WB	32.0	33.1	36.1	50.1	59.6	64.3	67.3	64.7	58.0	50.0	40.9	33.5	
		MCDB	33.0	34.2	39.1	58.6	68.5	72.7	74.9	72.4	64.0	52.7	42.0	34.1	
Mean Daily Temperature Range	5% DB	MDBR	9.3	12.1	13.5	17.9	20.1	18.3	18.1	19.4	17.0	12.4	8.2	8.4	
		MCDBR	6.6	7.9	17.8	25.5	26.5	25.1	23.6	25.3	24.3	19.9	11.3	6.7	
		MCWBR	6.6	7.0	11.6	13.1	12.2	11.9	10.5	12.1	12.7	12.9	9.7	6.5	
	5% WB	MCDBR	6.5	7.2	15.0	22.3	22.3	21.3	19.9	21.9	20.3	16.8	9.5	6.5	
		MCWBR	6.6	6.8	10.6	12.4	11.5	11.2	10.3	12.1	11.7	11.7	8.9	6.5	
Clear-Sky Solar Irradiance	taub		0.289	0.328	0.347	0.416	0.393	0.382	0.419	0.430	0.385	0.355	0.327	0.291	
	taud		2.122	2.052	2.106	2.126	2.302	2.354	2.270	2.226	2.326	2.353	2.239	2.129	
	Ebn at Noon		199	230	258	255	270	275	260	248	245	225	185	169	
	Edh at Noon		21	32	38	43	38	37	39	39	31	24	19	17	
All-Sky Solar Radiation	RadAvg		218	506	979	1282	1717	1789	1730	1436	920	439	174	121	
	RadStd		45	63	114	102	144	179	170	155	102	74	33	36	